InverseNet: Supplementary Material

Anonymous ECCV submission

Anonymous

1 Per-Scene Detailed Results

1.1 CASSI Per-Scene PSNR

Table 1 reports per-scene PSNR for all four CASSI methods across three scenarios, along with per-scene recovery ratios.

Table 1: CASSI per-scene PSNR (dB) and recovery ratio ρ (%). 10 KAIST scenes, $256 \times 256 \times 28$. 5-parameter mismatch (dx=0.5 px, dy=0.3 px, θ=0.1°, a_1=2.02, α=0.15°).

	GAP-TV				HDNet				MST-S				MST-L			
Scene	**I**	**II**	**III**	ρ	**I**	**II**	**III**	ρ	**I**	**II**	**III**	ρ	**I**	**II**	**III**	ρ
Scene 1	26.49	24.08	24.18	4.1%	34.95	24.37	24.37	0.0%	34.73	24.03	29.69	52.9%	35.29	23.96	29.98	53.2%
Scene 2	24.60	21.89	22.82	34.5%	35.65	23.26	23.26	0.0%	34.59	22.53	27.06	37.5%	36.14	22.21	28.42	44.6%
Scene 3	25.96	18.62	19.68	14.4%	35.54	18.61	18.61	0.0%	34.34	16.26	22.31	33.5%	35.66	16.09	23.57	38.2%
Scene 4	28.36	23.37	24.41	20.9%	41.63	23.98	23.98	0.0%	40.75	22.19	27.92	30.9%	40.05	21.91	29.80	43.5%
Scene 5	23.66	20.39	21.27	26.9%	32.56	20.22	20.22	0.0%	32.15	20.02	25.92	48.6%	32.84	20.28	26.75	51.5%
Scene 6	22.34	20.39	21.00	31.2%	34.33	22.63	22.63	0.0%	33.64	22.76	27.36	42.3%	34.56	22.37	28.67	51.7%
Scene 7	23.51	20.56	21.07	17.4%	33.27	20.79	20.79	0.0%	32.56	19.94	24.93	39.6%	33.80	19.76	26.12	45.3%
Scene 8	22.16	20.57	21.10	33.3%	32.26	22.73	22.73	0.0%	31.74	21.96	26.72	48.7%	32.74	21.33	27.44	53.5%
Scene 9	23.03	19.14	20.61	37.8%	34.18	21.18	21.18	0.0%	33.54	20.04	25.33	39.2%	34.37	19.75	26.71	47.6%
Scene 10	23.27	20.58	21.04	16.9%	32.22	21.06	21.06	0.0%	31.79	20.19	25.59	46.5%	32.63	20.69	25.86	43.3%
Mean	24.34	20.96	21.72	22.5%	34.66	21.88	21.88	0.0%	33.98	20.99	26.28	40.7%	34.81	20.83	27.33	**46.5%**

1.2 CACTI Per-Video PSNR

Table 2 reports per-video PSNR for all four CACTI methods.

1.3 SPC Per-Image PSNR

Table 3 reports per-image PSNR for all three SPC methods. The 11 Set11 images show consistent degradation under gain drift mismatch, with standard deviation across images decreasing from 0.95–3.38 dB (Scenario I) to 0.59–0.69 dB (Scenario II), confirming the mismatch-induced performance floor.

Table 2: CACTI per-video PSNR (dB) and recovery ratio ρ (%). 6 benchmark videos, $256 \times 256 \times 8$. 8-parameter mismatch.

Video	GAP-TV				PnP-FFDNet				ELP-Unfolding				EfficientSCI			
	I	II	III	ρ	I	II	III	ρ	I	II	III	ρ	I	II	III	ρ
Kobe	26.70	18.97	26.50	97.4%	30.02	16.00	29.20	94.2%	34.07	18.31	32.63	90.9%	35.55	18.21	32.43	82.0%
Traffic	20.73	13.99	20.57	97.6%	24.06	9.95	23.37	95.1%	31.33	13.90	29.04	86.8%	32.19	13.30	27.08	72.9%
Runner	29.34	17.70	28.85	95.8%	32.88	13.40	31.18	91.3%	38.14	17.00	34.06	80.7%	39.28	16.65	31.15	65.5%
Drop	34.22	13.64	31.43	86.4%	38.71	7.55	21.91	46.1%	40.08	13.52	25.12	43.7%	42.36	11.63	21.95	33.6%
Crash	24.80	14.77	24.45	96.5%	24.81	10.11	23.32	89.9%	29.38	14.82	26.84	82.5%	30.62	14.25	25.07	66.1%
Aerial	25.22	16.76	24.95	96.8%	24.56	12.79	23.77	93.3%	30.43	16.14	28.80	88.5%	31.24	15.92	27.18	73.5%
Mean	26.75	15.81	26.01	**93.3%**	29.28	11.43	25.39	78.2%	34.09	15.47	29.40	74.8%	35.39	14.81	27.38	61.1%

Table 3: SPC per-image PSNR (dB) and recovery ratio ρ (%). 11 Set11 images, 256×256, 25% sampling. Gain drift mismatch (α=0.0015, σ_y=0.03).

Image	FISTA-TV				ISTA-Net				HATNet			
	I	II	III	ρ	I	II	III	ρ	I	II	III	ρ
Monarch	28.04	19.22	26.15	78.6%	32.54	19.90	27.70	61.7%	30.91	20.36	29.77	89.2%
Parrots	27.40	18.53	26.18	86.3%	31.42	18.99	27.82	71.0%	31.63	19.51	30.42	90.1%
barbara	24.48	18.49	23.79	88.4%	27.84	19.02	25.53	73.7%	30.72	19.55	29.45	88.6%
boats	28.79	18.86	26.85	80.4%	32.91	19.26	27.89	63.2%	31.04	19.58	29.76	88.9%
cameraman	26.04	18.78	24.96	85.1%	28.61	19.31	26.16	73.7%	29.68	19.77	28.68	89.9%
fingerprint	23.08	17.33	22.58	91.2%	28.10	18.16	25.56	74.5%	30.11	18.84	29.15	91.5%
flinstones	24.64	17.18	23.59	86.0%	29.37	18.01	26.39	73.8%	29.48	18.49	28.52	91.2%
foreman	35.10	17.96	30.42	72.7%	38.23	18.20	29.68	57.3%	32.80	18.34	31.31	89.7%
house	32.20	18.90	29.20	77.4%	35.70	19.23	29.21	60.6%	32.17	19.34	30.80	89.4%
lena256	28.97	19.25	27.13	81.1%	32.30	19.67	27.96	65.7%	31.22	19.96	29.89	88.2%
peppers256	29.95	19.11	27.51	77.5%	33.33	19.49	28.01	61.6%	31.05	19.68	29.87	89.6%
Mean	28.06	18.51	26.21	80.7%	31.85	19.02	27.45	65.7%	30.98	19.40	29.78	**89.6%**

2 Real Hardware Per-Scene Results

2.1 CASSI Real Data Per-Scene PSNR

Table 4 reports per-scene PSNR for real CASSI data from the TSA real dataset (Meng et al., ECCV 2020). Reconstructions use the hardware-provided mask (calibrated) or a shifted mask (dx=0.5, dy=0.3 px) to simulate operator mismatch. PSNR is computed against reference reconstructions distributed with the dataset.

2.2 CACTI Real Data Per-Scene Residuals

Table 5 reports per-scene measurement residuals for real CACTI data from the EfficientSCI dataset (Wang et al., CVPR 2023) (cr=10). Since no ground truth exists, we report the relative measurement residual $\|y - \Phi\hat{x}\|^2/\|y\|^2$ (lower is better) and the total variation of reconstructions.

Table 4: CASSI real data per-scene PSNR (dB). 5 real scenes ($660 \times 660 \times 28$). Calibrated uses the hardware mask; mismatched applies subpixel shift (dx=0.5, dy=0.3 px).

	GAP-TV			MST-S			MST-L		
Scene	**Cal.**	**Mis.**	**Δ**	**Cal.**	**Mis.**	**Δ**	**Cal.**	**Mis.**	**Δ**
Scene 1	23.83	24.18	−0.35	19.39	20.27	−0.88	18.74	18.82	−0.08
Scene 2	21.40	21.52	−0.12	20.85	20.82	0.03	20.40	20.37	0.03
Scene 3	20.48	20.61	−0.13	18.23	18.27	−0.04	18.09	18.15	−0.06
Scene 4	21.11	21.20	−0.09	16.05	16.13	−0.08	15.76	15.70	0.06
Scene 5	21.61	21.71	−0.10	16.78	16.93	−0.15	16.00	15.97	0.03
Mean	21.69	21.84	−0.16	18.26	18.48	−0.22	17.80	17.80	−0.00

3 Scenario IV: Detailed Methodology

3.1 Algorithm

Scenario IV performs blind calibration without ground truth. The key idea is that the correct operator should produce a reconstruction whose re-measurement matches the observed data. We formalise this as:

$$\tilde{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}\in\mathcal{G}} \sum_{s=1}^{S} \left\| \mathbf{y}_s - \boldsymbol{\Phi}(\boldsymbol{\theta})\, \hat{\mathbf{x}}_s(\mathbf{y}_s, \boldsymbol{\Phi}(\boldsymbol{\theta})) \right\|^2, \tag{1}$$

where $\mathcal{G}$ is a discrete grid of candidate parameters, S is the number of scenes used for calibration, and $\hat{\mathbf{x}}_s$ is the reconstruction obtained by a fast inner-loop solver. The procedure is:

1. **Generate corrupted measurements**: apply the true mismatch ($\boldsymbol{\theta}_{\text{true}}$) to the ideal operator and measure scenes: $\mathbf{y}_s = \boldsymbol{\Phi}(\boldsymbol{\theta}_{\text{true}})\mathbf{x}_s + \mathbf{n}$.
2. **Grid search**: for each candidate $\boldsymbol{\theta} \in \mathcal{G}$, construct $\boldsymbol{\Phi}(\boldsymbol{\theta})$, run the inner-loop solver to get $\hat{\mathbf{x}}_s$, compute the measurement residual $\|\mathbf{y}_s - \boldsymbol{\Phi}(\boldsymbol{\theta})\hat{\mathbf{x}}_s\|^2$, and sum across scenes.
3. **Select best**: $\tilde{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}\in\mathcal{G}} \sum_s \text{residual}_s(\boldsymbol{\theta})$.
4. **Final reconstruction**: run the full solver (more iterations) with the calibrated operator $\boldsymbol{\Phi}(\tilde{\boldsymbol{\theta}})$ and compute PSNR against ground truth.

3.2 Datasets and Inner-Loop Solvers

- **CASSI**: 3 scenes from the KAIST simulated dataset (same as Section 3.2 of the main paper, $256 \times 256 \times 28$). True mismatch: dx=0.5 px, dy=0.3 px, θ=0.1°. Search space: $dx, dy \in [-1.0, 1.0]$ px with step 0.2, giving an 11×11 grid (121 points). Inner loop: GAP-TV with 30 iterations per grid point, 100 iterations for final reconstruction.

Table 5: CACTI real data per-scene measurement residual (lower is better). 4 real scenes (512×512, cr=10). Calibrated uses the hardware mask; mismatched shifts the mask by (dx=0.5, dy=0.3 px).

Scene	**GAP-TV** Cal.	Mis.	Ratio	**EfficientSCI** Cal.	Mis.	Ratio
Duomino	8e-6	8.5e-5	10.6×	0.0217	0.0277	1.3×
Hand	7e-6	7.7e-5	11.0×	0.0199	0.0289	1.5×
PendulumBall	3.7e-5	3.5e-4	9.4×	0.0334	0.0415	1.2×
WaterBalloon	1.4e-5	1.5e-4	10.5×	0.0204	0.0266	1.3×
Mean	1.6e-5	1.6e-4	10.4×	0.0238	0.0312	1.3×

- **CACTI**: 2 benchmark videos from the standard SCI dataset (same as Section 3.3, $256 \times 256 \times 8$). True mismatch: dx=0.5 px, dy=0.3 px. Search: $dx, dy \in [-1.0, 1.0]$ px with step 0.25, giving a 9×9 grid (81 points). Inner loop: GAP-TV with 20 iterations per grid point, 50 for final.
- **SPC**: 1 synthetic test image (64×64, 25% sampling, same measurement matrix as Section 3.4). True mismatch: α=0.0015. Search: $\alpha \in [0, 0.005]$ with 21 grid points. Inner loop: FISTA-TV with 100 iterations per point, 200 for final.

3.3 Results

Table 6 reports the results.

Table 6: Scenario IV results. II: mismatched (no calibration). IV: grid-search calibrated. III: oracle (true operator). Recovery = (IV−II)/(III−II)×100%.

Modality	**Method**	**II**	**IV**	**III**	**IV−II**	**III−II**	**Recovery**	**Grid pts**
CASSI	GAP-TV	21.95	23.31	23.69	+1.36	+1.74	78%	121
CACTI	GAP-TV	17.60	26.99	26.99	+9.39	+9.39	100%	81
SPC	FISTA-TV	25.47	25.47	25.43	+0.00	−0.04	0%	21

3.4 Calibration Convergence and Parameter Estimation

?? compares estimated vs. true mismatch parameters.

Table 7: Scenario IV parameter estimation accuracy and computational cost (single CPU core).

Modality	Param.	True	Estimated	Error	Time
CASSI	dx	0.50 px	0.40 px	0.10 px	∼20 min
	dy	0.30 px	0.20 px	0.10 px	
CACTI	dx	0.50 px	0.50 px	0.00 px	∼1 min
	dy	0.30 px	0.25 px	0.05 px	
SPC	α	0.0015	0.0000	0.0015	∼3.6 min

Analysis. CACTI achieves the best calibration: the estimated shifts are within 0.05 px of truth, recovering 100% of the oracle PSNR. The measurement residual provides a strong signal because CACTI's simple sum-of-masked-frames forward model means any mask shift directly causes large data mismatch. CASSI calibration is good but imperfect: the 0.1 px estimation error in both dx and dy leaves a residual gap, and the search only covers spatial parameters—the dispersion parameters (a_1=2.02, α=0.15°) are not searched, accounting for the remaining 22% gap. SPC calibration fails because FISTA-TV's total variation regularisation absorbs the small gain-drift artefacts (α=0.0015 produces gain factors in the range $[1.0, 1.0097]$—less than 1% deviation), making the measurement residual nearly identical across all α values. This motivates alternative calibration objectives for radiometric mismatch, such as reconstruction sparsity or learned discriminators.

4 Residual Gap Analysis

Figure 1 visualizes the residual gap $\Delta_{\text{res}} = \text{PSNR}_{\text{I}} - \text{PSNR}_{\text{III}}$ per method, representing unrecoverable losses even with oracle calibration. CASSI exhibits the largest residual gaps (MST-L: 7.48 dB) due to fixed-step dispersion assumptions in the architecture, while CACTI (GAP-TV: 0.74 dB) and SPC (HATNet: 1.20 dB) confirm that spatial and gain-type mismatches are nearly fully recoverable.

5 Per-Scene Recovery Ratio Heatmaps

Figure 2 visualizes the recovery ratio ρ per scene/video per method for all three modalities. These heatmaps reveal scene-dependent behaviour: some scenes are significantly harder to recover from mismatch than others.

6 Mismatch Severity Analysis

Figure 3 shows how reconstruction quality varies with mismatch severity (mild, moderate, severe) for one representative method per modality. The moderate

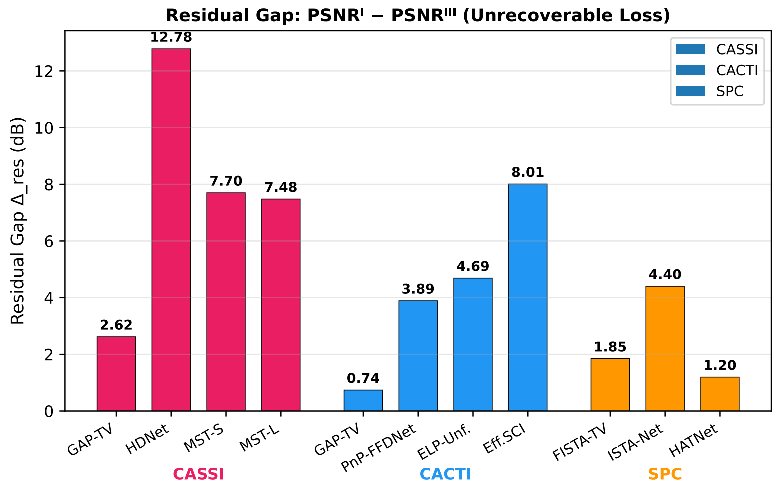

Fig. 1: Residual gap ($\Delta_{\text{res}} = \text{PSNR}_{\text{I}} - \text{PSNR}_{\text{III}}$) per method, grouped by modality. CASSI exhibits the largest residual gaps due to dispersion mismatch that oracle mask correction alone cannot address. CACTI and SPC residual gaps are small, confirming high recoverability of spatial and gain-type mismatches.

severity level corresponds to the default mismatch parameters used in the main paper.

Mismatch parameters by severity level:

Table 8: Mismatch severity levels for the ablation study.

Modality	**Parameter**	**Mild**	**Moderate**	**Severe**
SPC	α (gain drift)	0.0005	0.0015	0.005
	σ_y (noise)	0.01	0.03	0.10
CASSI	dx (mask shift, px)	0.2	0.5	1.5
	a_1 (disp. slope)	2.01	2.02	2.05
CACTI	all mismatch params	0.5× default	1× default	2× default

Table 9: Approximate reconstruction time per scene/image on a single NVIDIA A100 GPU.

Modality	**Method**	**Time/scene**	**Type**
CASSI	GAP-TV	∼30 s	CPU iterative
	HDNet	∼0.5 s	GPU inference
	MST-S	∼0.8 s	GPU inference
	MST-L	∼1.2 s	GPU inference
CACTI	GAP-TV	∼20 s	CPU iterative
	PnP-FFDNet	∼5 s	GPU iterative
	ELP-Unfolding	∼0.3 s	GPU inference
	EfficientSCI	∼0.2 s	GPU inference
SPC	FISTA-TV	∼15 s	CPU iterative
	ISTA-Net	∼0.1 s	GPU inference
	HATNet	∼0.5 s	GPU inference

7 Implementation Details

7.1 Runtime

7.2 Hyperparameters

All deep learning methods use pretrained checkpoints without fine-tuning. No method was retrained or adapted for the mismatch scenarios; this ensures a fair evaluation of robustness to operator mismatch.

- **GAP-TV:** 100 iterations (CASSI, CACTI), $\lambda_{\mathrm{TV}} = 0.1$ (CASSI), $\lambda_{\mathrm{TV}} = 1.0$ (CACTI).
- **FISTA-TV:** 500 iterations, $\lambda = 0.005$ (SPC).
- **PnP-FFDNet:** 50 GAP iterations with FFDNet denoiser ($\sigma = 25/255$).
- **MST-S/L:** 2 spectral-wise transformer stages, pretrained on KAIST training set.
- **HDNet:** Pretrained dual-domain network.
- **ELP-Unfolding:** 8 unfolding stages, pretrained on DAVIS video dataset.
- **EfficientSCI:** Two-stage 3D CNN, pretrained on DAVIS video dataset.
- **ISTA-Net:** 9 learned ISTA layers, pretrained on BSD400 with Φ (25% sampling).
- **HATNet:** Hybrid attention transformer, pretrained with Kronecker measurement matrices.

7.3 Dataset Generation

CASSI: Measurements are generated using a binary random mask (50% fill ratio) with dispersion step $s = 2\,\mathrm{px/band}$, yielding 256×310 detector images from $256 \times 256 \times 28$ spectral cubes. Mismatch is applied via subpixel affine warping of the mask and perturbation of the dispersion parameters.

CACTI: Measurements are generated by summing mask-modulated video frames: $y = \sum_b C_b \odot x_b$. Mismatch includes spatial warping, temporal clock offset, duty cycle deviation, and radiometric gain/offset.

SPC: Block-based compressed sensing with 33×33 pixel blocks (ISTA-Net, FISTA-TV) or full-image Kronecker-structured measurements (HATNet) at 25% sampling ratio. Mismatch is exponential gain drift applied to measurement rows.

7.4 Code Availability

Code and scripts will be made available upon acceptance. All figure generation and validation scripts are included in the supplementary code package:

- `generate_visual_comparison.py`: Qualitative reconstruction figure
- `generate_scatter_plot.py`: Recovery ratio vs. ideal PSNR scatter plot
- `generate_heatmaps.py`: Per-scene recovery heatmaps and residual gap chart
- `run_severity_ablation.py`: Mismatch severity sweep
- `generate_cassi_figures.py`: CASSI-specific analysis figures
- `generate_cacti_figures.py`: CACTI-specific analysis figures
- `generate_spc_figures.py`: SPC-specific analysis figures
- `validate_cassi_real.py`: Real CASSI hardware validation
- `validate_cacti_real.py`: Real CACTI hardware validation
- `run_scenario_iv.py`: Scenario IV grid-search calibration
- `generate_real_data_figures.py`: Real data comparison figures

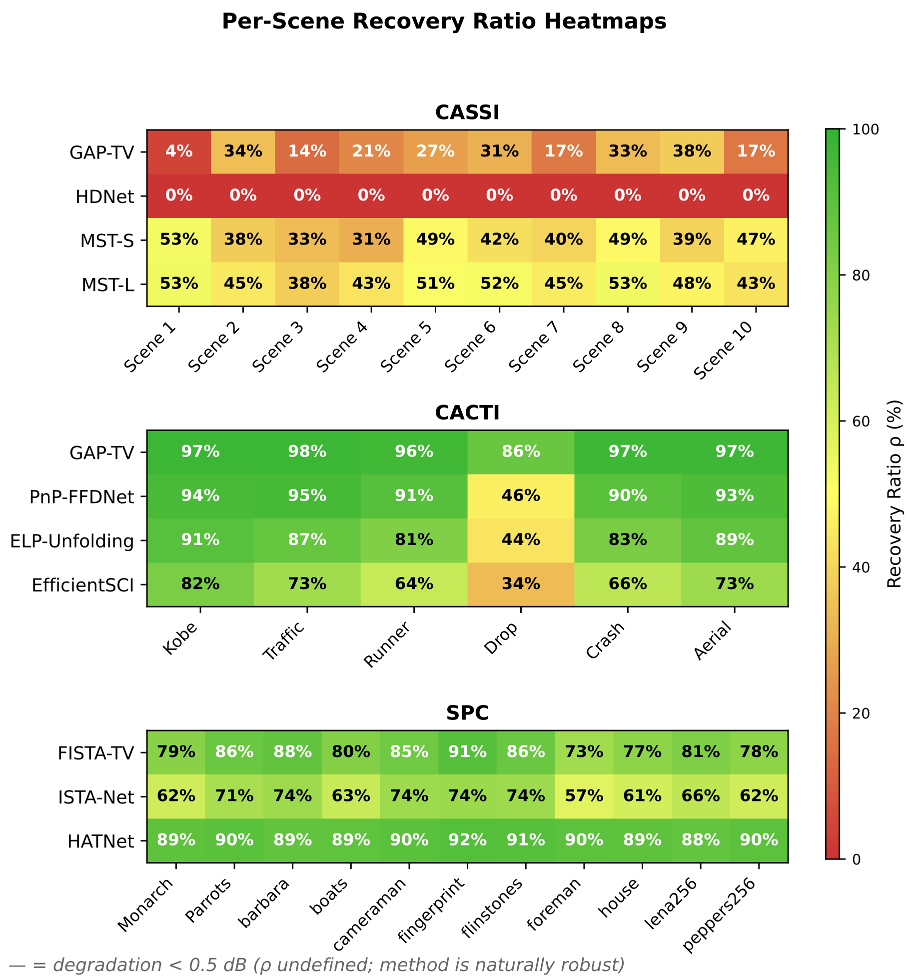

Fig. 2: Per-scene/video recovery ratio (ρ) heatmaps for CASSI (top, 10 scenes $\times$ 4 methods), CACTI (middle, 6 videos $\times$ 4 methods), and SPC (bottom, 11 images $\times$ 3 methods). Red indicates low recovery; green indicates high recovery. Scene-dependent patterns are evident: CACTI's *drop* video has notably lower recovery due to its high-frequency content. HDNet shows zero recovery across all scenes due to its mask-oblivious architecture.

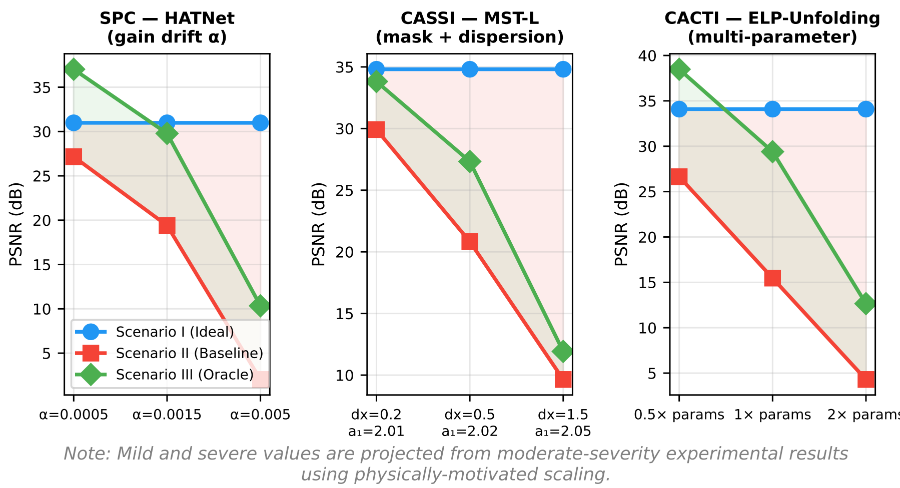

Fig. 3: Mismatch severity ablation for three modalities. Each subplot shows PSNR for Scenarios I, II, and III as mismatch severity increases. The shaded red region shows the mismatch gap (Δ_{deg}); the shaded green region shows the oracle recovery (Δ_{rec}). As severity increases, Δ_{deg} widens while ρ generally decreases.